Array

- **Array** is a collection of items of the same Objects that are stored at contiguous memory locations.

- **Array Index:** In an array, Objects are identified by their indexes. Array index typically starts from 0.

- **Array Length:** The length of an array is determined by the number of Objects it has.

Memory Representation of Array

Memory location	1000	1004	1008	1012	1016	1020	1024	2028	1032	1036
Object	17	25	110	53	66	7	77	31	65	119
Index	0	1	2	3	4	5	6	7	8	9

Array Manipulation

```
// Create a student list

//inputs of student id, name, and subjects

// Calculate average of marks, and then display student id, name, marks , avg.

// Insert an element at a specific location

// Delete an element from the array

// Search for a target element
```

Time Complexity

Operation	Best Case	Average Case	Worst Case
Traversal	O(N)	O(N)	O((N)
Insertion	O(1)	O(N)	O(N)
Deletion	O(1)	O(N)	O(N)
Searching	O(1)	O(N)	O(N)

Auxiliary Space Complexity

Operation	Best Case	Average Case	Worst Case
Traversal	O(1)	O(1)	O(1)
Insertion	O(1)	O(N)	O(N)
Deletion	O(1)	O(N)	O(N)
Searching	O(1)	O(1)	O(1)